CSE STUDY HUB

# Digital Forensics & Incident Response

UNIT 6 · Research Innovations and Advanced Case Studies

## What this unit is about

This unit builds understanding of **Research Innovations and Advanced Case Studies**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

### Syllabus coverage checklist

**1.** Emerging areas: cloud, IoT, vehicle, drone and blockchain forensics

**2.** Challenges: encryption, scale, jurisdiction, anti-forensics and volatile evidence

**3.** Case-study method: scenario, evidence sources, timeline and hypotheses

**4.** Build an evidence matrix linking artefacts to claims

**5.** Evaluate tools: repeatability, validation, accuracy and limitations

**6.** Present findings with clear methods, hashes, exhibits and cautious conclusions

**7.** Research question, literature review, method, results and future scope

**How to study:** Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

### Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

# UNIT 6 NOTES · PART 1

## Emerging areas: cloud, IoT, vehicle, drone and blockchain forensics

Cloud computing provides on-demand shared computing resources over a network. IaaS supplies infrastructure, PaaS supplies a development platform, and SaaS supplies a ready-to-use application.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Challenges: encryption, scale, jurisdiction, anti-forensics and volatile evidence

Volatile evidence can disappear when power is lost or a system changes. RAM, running processes, active network connections and logged-in sessions should generally be considered before persistent disk data.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Case-study method: scenario, evidence sources, timeline and hypotheses

Present Case-study method: scenario, evidence sources, timeline and hypotheses as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Connect and apply

For a practical question on Research Innovations and Advanced Case Studies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

# UNIT 6 NOTES · PART 2

## Build an evidence matrix linking artefacts to claims

Build an evidence matrix linking artefacts to claims is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Evaluate tools: repeatability, validation, accuracy and limitations

Compare the alternatives in Evaluate tools: repeatability, validation, accuracy and limitations using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Connect and apply

For a practical question on Research Innovations and Advanced Case Studies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

# UNIT 6 NOTES · PART 3

## Present findings with clear methods, hashes, exhibits and cautious conclusions

A hash function maps a key to a table index. Collisions are handled by chaining or open addressing. Performance depends on a good hash function, load factor and resizing policy.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Research question, literature review, method, results and future scope

Present Research question, literature review, method, results and future scope as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

**Write in your own words:** Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

## Connect and apply

For a practical question on Research Innovations and Advanced Case Studies, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

# QUICK REVISION & EXAM PRACTICE

## Key takeaways

• **Emerging areas: cloud, IoT, vehicle, drone and blockchain forensics** — know the definition, core working and one example.

• **Challenges: encryption, scale, jurisdiction, anti-forensics and volatile evidence** — know the definition, core working and one example.

• **Case-study method: scenario, evidence sources, timeline and hypotheses** — know the definition, core working and one example.

• **Build an evidence matrix linking artefacts to claims** — know the definition, core working and one example.

• **Evaluate tools: repeatability, validation, accuracy and limitations** — know the definition, core working and one example.

• **Present findings with clear methods, hashes, exhibits and cautious conclusions** — know the definition, core working and one example.

• **Research question, literature review, method, results and future scope** — know the definition, core working and one example.

## Worked-answer framework

**Scenario:** A question asks you to explain or apply a concept from Research Innovations and Advanced Case Studies. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

## Self-test questions

**Q1.** Define Emerging areas: cloud, IoT, vehicle, drone and blockchain forensics and explain its importance.

**Q2.** Explain the working/steps involved in Challenges: encryption, scale, jurisdiction, anti-forensics and volatile evidence.

**Q3.** Differentiate between two related concepts from Research Innovations and Advanced Case Studies.

**Q4.** Give one practical application of Research question, literature review, method, results and future scope and discuss one limitation.

**Q5.** Write a structured 5-mark answer covering the main ideas of Research Innovations and Advanced Case Studies.

**Final checklist:** I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.